

To-do workflow

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|                               🚀 STORY-DRIVEN EXPERIMENT WORKFLOW / TIMELINE
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Note:
[0] Our (no-skip) = A heavy classifier trained with million images ("heavy but safe")
[1] M1 = A lightweight image classifier ("fast but unsafe")
[2] M2 = A lightweight object detector; ("fast but unsafe")
[3] M3 = A temporal voting method (video-level baseline) ("accurate but delay to detection")

The emojis make it instantly scannable: 🗳️ for decisions, ✅/❌ for pass/fail, 🕒 for recall/safety, ⚡ for speed, etc.—while preserving the exact sequential dependencies from the table and paper outline

🔍 [^0] DATASET CHECK ✅
|
|-- verify split and balance:
|   Train/Val = 90 videos 📊
|   Test      = 60 videos 📊
|   Classes   = FireSmoke / Safe
|
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⚡ [^1] TEST-SET BASELINES FIRST 🚩
|
|-- Run BIG MODEL (no skip) on TEST
|   -> get reference accuracy / recall / latency 🕒
|
|-- Run M1 on TEST
|   -> get lightweight speed / recall tradeoff 🕒⚡
|
|-- Run M2 on TEST
|   -> get lightweight speed / recall tradeoff 🕒⚡
|
+--> 🗳️ Decision A:
    "Do we clearly see:
    BIG MODEL = accurate but slow 🕒,
    M1/M2 = faster but miss anomalies ❌?"
    |
    +-- ❌ NO --> 🔗 re-check data / labels / eval pipeline
    |
    +-- ✅ YES --> continue
        |
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📊 [^2] DEFINE METRICS + SAFETY RULE 🛡️
|
|-- Recall(Anomaly) 🕒
|-- Filter Rate on Safe frames 🕒
|-- System FAR
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|-- Safety constraint: Recall >= 99%
|
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🔧 [^3] VALIDATION-SET TUNING 🛠️
|
|-- Approach 1: naive motion 🦾
|-- Approach 2: motion + color + texture 🧐👤🗣️
|-- Grid search on VAL only
|
+--> 🕒 Decision B:
    "Is there a config satisfying Recall >= 99%? 🛡️"
    |
    +-- ✗ NO --> 🔧 revise thresholds / rules
    |
    +-- ✔️ YES --> choose best config by SkipScore ☆
        |
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✔️ [^4] COMPONENT TEST ON TEST SET 🛠️
|
|-- Run final Approach 1 alone
|-- Run final Approach 2 alone
|
|-- Measure:
|   Recall(Anomaly) 🕒
|   Filter Rate (Safe) 🕒
|   Overhead (ms) 🕒
|
+--> 🕒 Decision C:
    "Does Approach 2 behave like a safe gatekeeper? 🛡️"
    |
    +-- ✗ NO --> go back to [^3] 🔄
    |
    +-- ✔️ YES --> continue
        |
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🚀 [^5] FULL PIPELINE TEST ON TEST SET 🛠️
|
|-- Approach 1 + BIG MODEL
|-- Approach 2 + BIG MODEL
|-- Compare against:
|   BIG MODEL only
|   M1
|   M2
|
|-- Measure:
|   system recall / F1 🕒
|   latency 🕒
|   FPS ⚡
|   compute reduction 📉
|
+--> 🕒 Core Claim Check:
    "Does Approach 2 + BIG keep BIG MODEL accuracy
    while reducing compute and latency? 🦾"
    |
    +-- ✗ NO --> go back to [^3] or [^4] 🔄
    |
    +-- ✔️ YES --> continue
        |

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🕒 [^6] TEMPORAL BASELINE COMPARISON 🏆
|
|-- Compare against M3
|-- Measure:
|   time-to-first-alarm 🚨⚡
|   video-level recall
|   false alarm rate
|
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📊 [^7] ABLATION + QUALITATIVE EVIDENCE 📷
|
|-- motion only
|-- + fire rules 🔥
|-- + smoke rules 🚬
|-- full rules
|
|-- collect edge cases:
|   smoke missed by Appr.1 ✖
|   safe motion falsely triggered by Appr.1 ⚠
|
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⚠ [^8] LIMITATION CHECK 🛠
|
|-- low light / IR 🌙
|-- fog / weather 🌫
|-- unusual lighting
|
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🏆 +-----+
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| FINAL STORY: redundancy is the bottleneck; a safe skip module lets BIG MODEL
|
| run much faster without losing safety-critical recall. ⚡🔧
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Compact paper version (with emojis)

Perfect for pasting into manuscripts or presentations:

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|
| 🚀 PAPER2 TARGET TABLES WORKFLOW (All-in-One)
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🔍 [Baselines + Component Test] → Table 1 ✅
|
+-- Table 1: Standalone Recall and Filtering Performance
|
| | Method | Overhead | Recall | Filter Rate |
| |-----|-----|-----|-----|
| | M1 (MobileNetV2 Classifier) | 15.0ms | 85.3% | N/A |
| | M2 (YOLOv8-Nano Detector) | 22.0ms | 88.2% | N/A |
| | Appr.1 (Naive Motion) | 1.2ms | 97.2% | 65.0% |
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|      | Appr.2 (Rule-Based)          | 2.5ms   | 99.1% | 72.1% |
|
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🔍 [VAL Grid Search] → Tables 5&6 ☆
|
+-- Table 5: Hyperparam Search Space (Target: 16x16, 1.5%, smoke rules)
|      | Param                      | Appr.1 | Appr.2 |
|      |-----|-----|-----|
|      | Grid Size (NxN)              | 16x16  | 16x16  |
|      | Motion Threshold             | 2%     | 1.5%   |
|      | Grayness Threshold (Smoke)   | N/A    | [0.7,0.95]|
|
+-- Table 6: Top 3 Configs (Val Set)
|      | Rank | Config                                | Recall | Filter | SkipScore |
|      |-----|-----|-----|-----|-----|
|      | 1    | 16x16,1.5%,Smoke Rules              | 99.1%  | 72.1%  | 0.714     | ✓
Appr.2
|      | 2    | 16x16,2.0%,Smoke Rules              | 99.0%  | 68.3%  | 0.676     | ✗
|      | 3    | 32x32,1.0%,Smoke Rules              | 99.2%  | 65.7%  | 0.652     | ✗
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🚀 [Full Pipeline Test] → Table 2 📄
|
+-- Table 2: End-to-End System Performance
|      | Config                        | Latency | FPS   | Recall | Cost Red. |
|      |-----|-----|-----|-----|-----|
|      | BIG MODEL only                | 50.0ms  | 20    | 98.5%  | 0%         |
|      | M1                             | 15.0ms  | 66    | 76.0%  | 70%        |
|      | M2                             | 22.0ms  | 45    | 81.5%  | 56%        |
|      | Ours (Appr.1 + BIG)           | 18.3ms  | 54    | 98.3%  | 63%        |
|      | Ours (Appr.2 + BIG)           | 16.5ms  | 60    | 98.5%  | 67%        |
|
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🏆 [vs M3 Temporal] → Tables 3&4 ⌚
|
+-- Table 3: Time-to-Alarm Analysis (*M3 using BIG Model*)
|      | Method                | Scope    | Delay   | Min Latency | First Alarm |
|      |-----|-----|-----|-----|-----|
-|      | M3 Temporal Voting    | Video(30f) | 25ms/f  | 30 Frames   | >750ms
|
|      | Ours (Appr.2+BIG)     | Frame-level| 16.5ms/f| 1 Frame     | ~52.5ms
|
✓
|
+-- Table 4: Video-Level Metrics
|      | Method                | Video Recall| False Alarm | Compute/Sec |
|      |-----|-----|-----|-----|
|      | M3 Temporal Voting    | 97.5%      | 1.2%       | 750ms       |
|      | Ours (Appr.2+BIG)     | 98.4%      | 1.5%       | 495ms       | ✓
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📊 [Ablation + Limits] → Qualitative Figure 4 📈⚠️

📖 STORY: BIG MODEL accuracy ⚡ 3x faster with safe skip module!
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